

Summary of Change - Matchmaking/Leaderboard

We introduced an enum for game types, which was not originally planned. Up until last week, the game type was specified using an array of numbers, which initially worked well. However, maintaining a consistent format across the codebase became increasingly difficult. To address this, we decided to implement an enum for gameType to improve readability and maintainability.

One of the reviewing team members recommended implementing an ELO rating system for player rankings. After careful consideration, we chose to implement a **Matchmaking Rating (MMR)** system instead. This decision was based on MMR's flexibility and its ability to account for individual performance metrics beyond simple win/loss outcomes.

Our implementation uses individual win ratios as a proxy for player skill and dynamically adjusts MMR based on:

- Match outcome (win/loss/draw)
- Relative performance between players
- Game type (TicTacToe, Connect4, or Checkers)

We also updated the planning documents, added a proper timeline to track and manage work, and included details about the classes implemented for the Matchmaking-Leaderboard system. A brief overview of the ranking system and its functionality was also provided to help ensure clarity and alignment within the team.